LAB 1: FUNDAMENTAL PARAMETERS OF ANTENNAS

1. Objectives and preliminary considerations

The first objective of this lab is to review the fundamental parameters of antennas covered in class. The second objective is to carry out a tutorial of FEKO, an electromagnetic simulation software (http://www.feko.info).

Each group must submit a report of the work carried out during the lab. It is highly recommended to bring a pen drive to store data/figures that you may use later in the preparation of the report. After the session there is no guarantee that this information remains on your computer. The deadline for delivering the report is March 30th.

Prior to the start of the lab you must also download the document GetStarted.pdf from Campus Virtual and read the first and second sections. You can use this document during the laboratory session.

2. Practical session

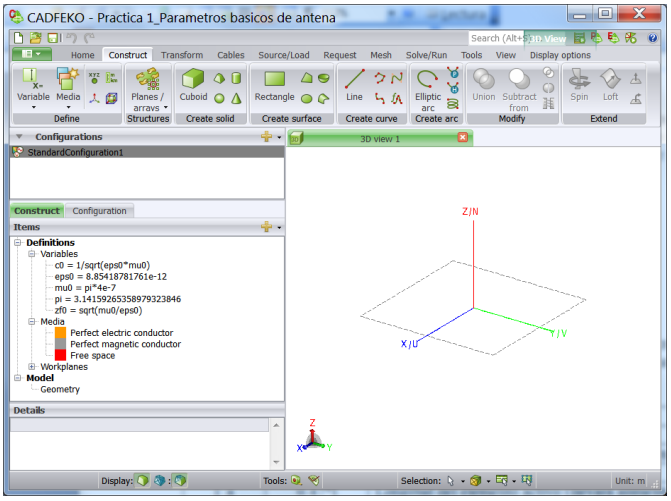
This session is intended to make a short tutorial introducing some of the functions of FEKO. Follow the instructions of the text. The report will consist of answering the questions asked throughout the tutorial.

We are going to create a simplification of a Yagi antenna (commonly used for television reception) with the following features:

- Operating frequency: 400 MHz (λ = 0.75 m).
- 15 elements: 13 directors, one reflector and one driven (energized) element.

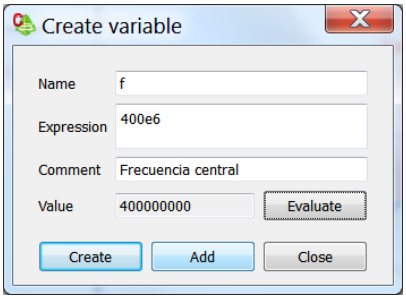
2.1. Creating an antenna model with CADFEKO

CADFEKO is the module of FEKO that allows to create the structure to be analysed and establish the simulation parameters. Start the application by double-clicking the icon CADFEKO (Suite 7.0) on the desktop. In the window that opens select the option *Create a new model* and the CAD environment will appear.



2.1.1. Definition of variables

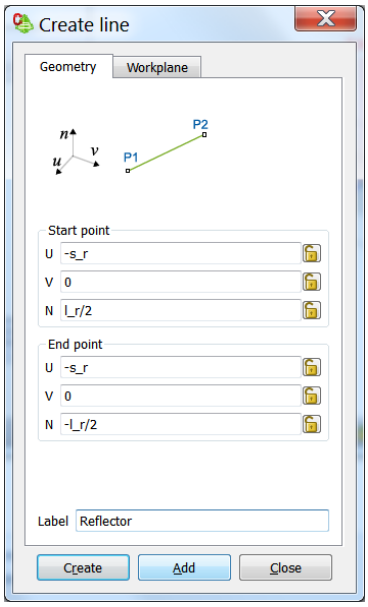
Let's start by defining a set of variables that will facilitate the creation of the structure and will allow us to make changes efficiently. Select *Construct >> Variable >> Add Variable* in the top menu.



Name	Expression	Comment (you needn't to copy them)
f	400e6	Operating frequency
l	c0/f	Wavelength (you can use equations)
l_r	0.5*l	Reflector length
l_a	0.47*l	Driven element length (energized)
l_d	0.406*l	Directors length
s_r	0.25*l	Spacing between driven element and reflector
s_d	0.34*l	Spacing between driven element and director Spacing between directors
a	0.003*l	Wire radius

2.1.2. Creating the antenna

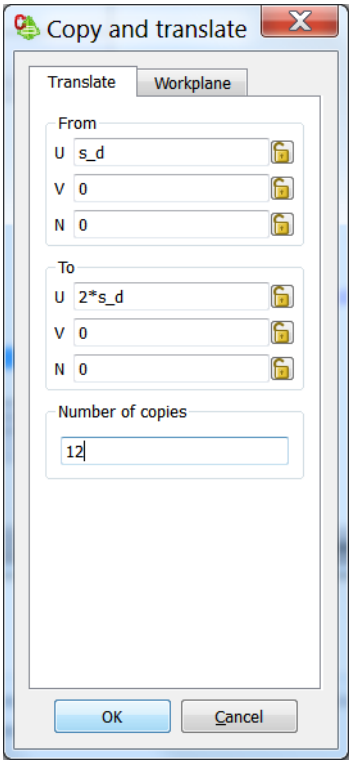
We start by creating 3 wires: the reflector, the driven element and one of the directors. Select *Construct>>Line* in the top menu (repeat the steps for each wire)

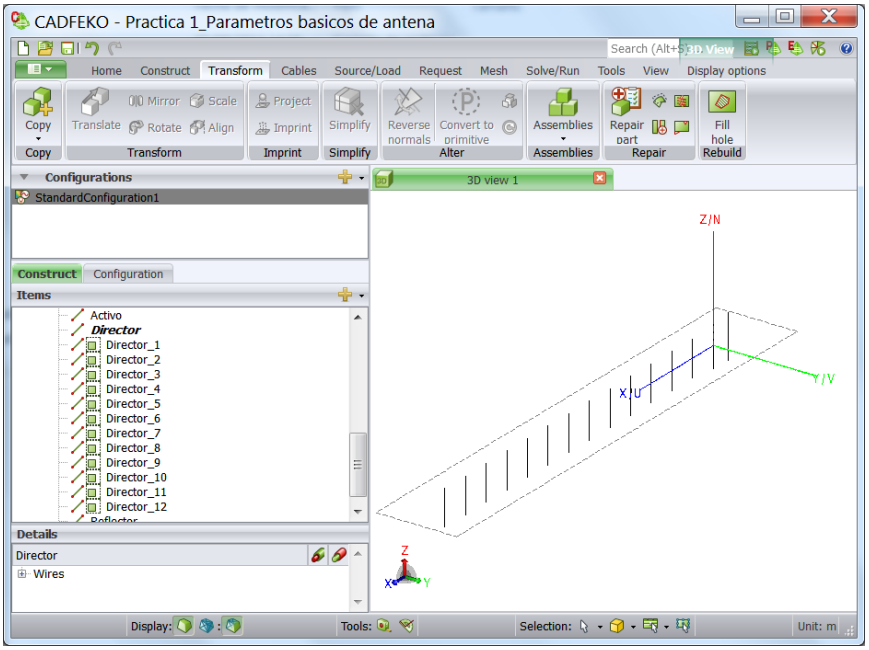


Label	Start point			End point		
	U	V	N	U	V	N
Reflector	-s_r	0	l_r/2	-s_r	0	-l_r/2
Driven	0	0	l_a/2	0	0	-l_a/2
Director	s_d	0	l_d/2	s_d	0	-l_d/2

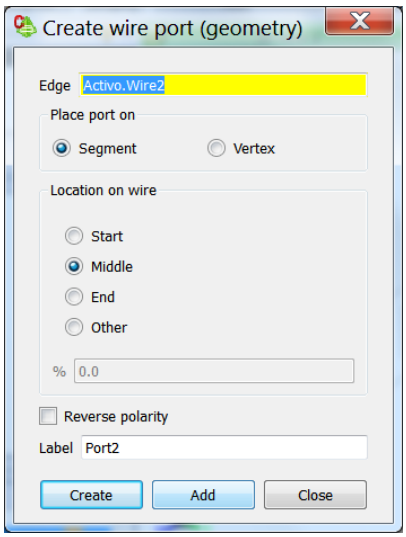
In order to add the remaining 12 directors, you can either repeat the previous step using the spacing between directors or use the copy and translate option: select *Director* in the left model tree on the left side (*Construct/Items/Model/Geometry*) and then *Transform>>Copy special…>> ... and translate* in the top menu. Fill the new window with the following information.

Number of copies	From			To		
	U	V	N	U	V	N
12	s_d	0	0	2*s_d	0	0



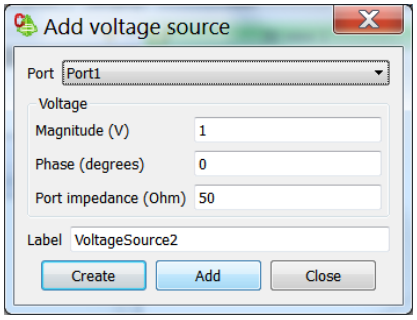


2.1.3. Feeding the antenna



Now, we will feed the antenna by positioning a port. Select *Driven* in the model tree on the left side (*Construct/Items/Model/Geometry*). In the lower corner open the details of *Wires* and select the *Wire* that appears. Now, press the right mouse button and then *Create Port >> Wire port*. In the window that opens select the location *Middle* (of the wire), then press *Add* and subsequently *Close*.

You'll see as a new element *Port 1* has just appeared in the model tree (*Construct*). Select that element, then press the right mouse button and select *Voltage Source.* This will open a new window that will allow us to assign a source of 1 V and 50 Ω to *Port 1*. Press *Add* and then *Close*. As a result the model tree will change to the tab *Configuration* and a *Voltage Source* will be hanging from *Sources*.



Save the model by clicking on the icon with a diskette that appears in the top left corner. Repeat this operation whenever you consider it convenient.

2.2. Analysis of the antenna as a circuital element

2.2.1. Simulation settings

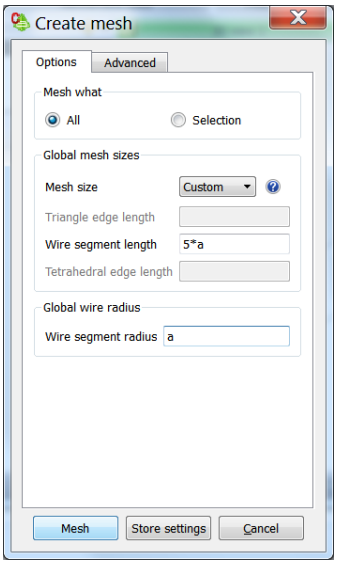
Select the tab *Configuration* in the model tree on the left side if it is not already selected. By double clicking in *Global >> Frequency*, a new window will open to allow you to specify the frequency range to be analyzed. In the tab *Frequency* select the option *Continuous (interpolated) range* and specify *Start frequency (Hz)* 0.9*f and *End frequency (Hz)* 1.1*f. In this option FEKO chooses the simulation frequencies according to an algorithm and interpolates the results with a curve. The final result is a smoother response although sometimes the algorithm has problems to converge. In order to minimize these problems select the tab *Advanced* and follow the steps below:

- Check the box *Specify maximum number of samples* and enter the value 21. FEKO will stop the simulation if it has not converged after 21 frequencies and will provide the approximation at that point.
- Uncheck all the options of *Quantities to include for adaptive frequency sampling* except *Impedance data*. Thus, FEKO will only use this information in its algorithm.

Now, press button *Ok* to accept and close the window.

2.2.2. Meshing the antenna and simulating

The next step is to decompose the antenna in small elements that approximate the currents on it. This process is called meshing the antenna. If we have a wire we get segments, in the case of surfaces we get triangles and for a volume we get tetrahedral. A fine mesh (small elements) usually leads to more accurate results, however, the computation time increases significantly with the number of elements. For accurate results the maximum size of the elements should be 10 to 20 times smaller than the wavelength of the highest frequency to simulate.



In the top menu select *Mesh >> Create Mesh*. Then, select *Mesh size >> Custom* in the new window and specify:

- *Wire segment length*: 5*a
- *Wire segment radius*: a

In order to generate the mesh press *Mesh*. In the message window you will find the number of segments in which each part of the antenna has been decomposed.

Next, select *Solve/Run >> Feko Solver* in the top menu to run the component of FEKO that carries out the electromagnetic simulation (RUNFEKO). Save changes before starting the simulation. Once the simulation is over press *Ok* to close the window of RUNFEKO.

2.2.3. Analysis of results

POSTFEKO is the component of FEKO that displays the simulation results. In the top menu of CADFEKO select *Solve/Run >> Postfeko* in order to launch this component with the information of the last simulation. A new window with certain similarities with CADFEKO will open, in the middle you will find the simulated antenna.

To create a new graph in Cartesian coordinates, select *Home >> Cartesian* in the top menu. The information about impedance/matching can be added by selecting *Home >> Source Data >> Voltage Source*. Another option is selecting and dragging to the new graph the *Voltage Source* that appears in the tree on the left.

In the menu on the right you can see that the magnitude of the reflection coefficient is being represented and that 50 Ω impedance has been used as the reference for the computation. Select the logarithmic scale (*dB*). In the lower right corner, you can activate cursors that may help you to make measurements.

P1. At what frequency the first maximum matching occurs? What is the reflection coefficient at that frequency? If the bandwidth is defined for a return loss larger than 10 dB find the absolute and relative bandwidth.

P2. Answer again the previous questions if the reference impedance is 75 Ω.

Now, create a new graph in Cartesian coordinates and add *Voltage Source* twice. In the right menu two items appear. Use the first one (in blue) to plot the real part of the input impedance (*Quantity: Impedance, Real*) and the other one (in green) to plot the imaginary part of the input impedance (*Quantity: Impedance, Imaginary*).

P3. Determine the input impedance at the ends of the band and at the frequency of maximum matching. Comment the results.

In a third graph in Cartesian coordinates add *Voltage Source* and plot the voltage standing wave ratio (*Quantity: VSWR 1:10*).

P4. Which is the range of values of the VSWR for a return loss larger than 10 dB for a reference impedance of 50 Ω?

Open a new graph with the Smith chart (*Home >> Smith*) and plot the input impedance.

P5. In what area of the Smith Chart (in one side, up, down..., comment the results) you can find the frequencies with higher return loss (better matching)?

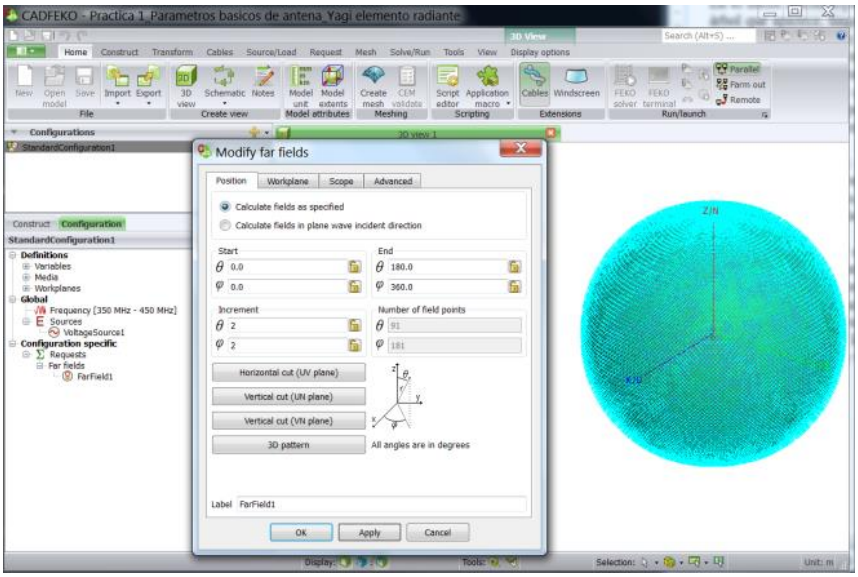
Next, we will check that the antenna efficiency is 100% for all the frequencies because it has been built using perfect electric conductors. Add a graph in Cartesian coordinates and a result of power (*Power)*. Then, in the right, select *Efficiency* in the drop-down menu of *Quantity*.

Finally, close POSTFEKO saving, if you wish, the results in a file.

2.3. Analysis of the antenna as a radiating element

2.3.1. Simulation settings

Select the tab *Configuration* in the left side of CADFEKO. In the tree that appears, click the right mouse button on *Request*. In the pull-down menu select the option *Far Field*, a new window will open. Press the *3D Pattern* button and some boxes will be filled automatically in order to obtain the radiated fields in a sphere that encloses the antenna. In order to increase the density of sampling directions (marked with “x”) enter an *Increment* of 2º in elevation and azimuth. Press *Create* to confirm and close the window.



Now double click on the option *Frequency* that can be found in the tree of the tag *Configuration*. In the window that opens select the tag *Advanced* and include *Far Field* in the quantities to consider for carrying out the frequency adaptive sampling. Press *Ok* to accept.

Since the antenna has not been modified it is not necessary to mesh again before starting the simulation. In the top menu select *Solve/Run >> Feko Solver* to run RUNFEKO. Save changes before starting the simulation. Now the simulation will take longer (more frequencies), in fact, it will end without achieving convergence. As a consequence, results could not be reliable, but we are going to consider that they are correct. Once the simulation is finished, press *Ok* to close the window with RUNFEKO.

2.3.2. Analysis of results

In the top menu of CADFEKO select *Solve/Run >> Postfeko* in order to launch this component. Create a graph in Cartesian coordinates and plot the reflection coefficient (*Voltage Source*) in order to have it as a reference. In another Cartesian graph add a measure of *Far Field*. You will find as the antenna gain (*Quantity: Gain*) in direction *Theta = 0, Phi = 0* is zero. In order to get the gain in the direction of maximum radiation select *Theta = 90, Phi = 0* in the right menu, then, activate the logarithmic scale.

P6. Which is the frequency of maximum gain? What is the maximum gain of the antenna (in dB)? If the gain bandwidth is defined for a reduction of 3 dB of the maximum value, find the absolute and relative bandwidth.

Now, we are going to add two new measurements of *Far Field*: press the corresponding button in the top menu or, alternatively, select the trace *Far Field1* in the right menu, click with the right mouse button and then select *Duplicate trace (Ctrl+k)*. Plot directivity (*Quantity: Directivity*) and realised gain (*Quantity: Realised Gain*) in the second and third trace respectively. The realised gain includes the effects of mismatching in the gain. Make sure that both traces show the direction of maximum radiation and that the scale is logarithmic.

P7. Why gain and directivity are equal? Why realised gain is always smaller than gain? At what frequency gain and realised gain are approximately equal? Why?

Next, we are going to focus in the analysis of the radiation pattern at 400 MHz. Select the graph where you can see the 3D view of the antenna and add a *Far Field* result. Then, select the frequency of 400 MHz in the right menu. You can rotate the radiation pattern clicking on the 3D view and dragging the mouse around. By default, the radiation pattern is in linear scale. If you select the logarithmic scale the side lobes are better appreciated, however you should take into consideration the following aspects:

- *The radiation pattern represents the antenna radiation properties in the far field (large distances). Therefore, when we observe the radiation pattern, the antenna is just a small point in the coordinate origin. Although in FEKO the antenna protrudes from the radiation pattern, this is not real, FEKO uses this representation to show the relative position of the antenna with respect to the radiation pattern.*

- *The appearance of a given radiation pattern depends strongly on the dynamic range used in the logarithmic representation. Select Display >> Individual Range in the top menu. Then, in the new window, choose to specify the dynamic margin of the logarithmic scale. Try with 20 dB and 100 dB, you will see as the appearance changes significantly.*

In order to extract more details of our radiation pattern we are going to analyse the E-plane and H-plane cuts. Open a new window in polar coordinates (top menu *Home >> Polar*) and add a *Far Field* at 400 MHz. The E-plane cut corresponds to the XZ plane (*Independent axis: Theta (wrapped), Phi = 0*), activate the logarithmic scale. Now select the trace in the right menu and press *Ctrl+k* to create a copy. The H-plane cut corresponds to the XY plane (*Independent axis: Phi, Theta = 90*). Remember that the antenna has no losses, so directivity and gain coincide.

P8. Determine, for each cut, the maximum radiation, the 3 dB beamwidth, the side lobe ratio (SLR) and the front to back ratio (FBR). You can use cursors and/or normalize the representations.

P9. Estimate the antenna directivity using the obtained 3 dB beamwidth. Try to justify differences between the estimated and measured value.

P10. Look on the Internet for a real Yagi antenna and shown anything that may be of interest to you.